Chapitre 1 : Le langage R

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1 Avant de commencer

Ce chapitre reprends les codes du chapitre 1 sur ***le langage R*** du livre **Langage R et Statistiques** : *Initiation à l'analyse de données* publié par les éditions ENI (2ème édition).

```
tibble::view(women)
```

2 Installation de R et RStudio

2.1 installation de l'addin `{esquisse}`

```
install.packages("esquisse")
```

2.2 installation du `{tidyverse}`

```
install.packages("tidyverse")
```

3 Fonctions et package

3.1 utilisation du préfixe

```
# ne fonctionne pas
data.table(
  x = seq(from = 1, to = 50, by = 5)
)
```

```
Error in `data.table()`:
! impossible de trouver la fonction "data.table"
```

```
# fonctionne
data.table::data.table(
  x = seq(from = 1, to = 50, by = 5)
)
```

```
        x
    <num>
 1:     1
 2:     6
 3:    11
 4:    16
 5:    21
 6:    26
```

```
 7:    31
 8:    36
 9:    41
10:    46
```

3.2 affichage de la page d'aide

```
?seq
```

4 Utilisation du format projet

4.1 Création d'un dossier data

```
dir.create("data")
```

5 Initiation au codage en R

5.1 Nomenclature du langage R

5.1.a Calcul d'une somme

on calcule la somme de 3+8

```
3 + 8
```

```
[1] 11
```

5.1.b Création d'un objet

```
ceci_est_un_objet <- 4.5 + 1.1
```

afficher un objet dans la console en utilisant son nom

```
ceci_est_un_objet
```

```
[1] 5.6
```

il est possible de faire de l'utiliser

```
ceci_est_un_objet * 2
```

```
[1] 11.2
```

mais cela ne modifie pas l'objet

```
ceci_est_un_objet
```

```
[1] 5.6
```

5.1.c Modification d'un objet

```
ceci_est_un_objet <- ceci_est_un_objet * 2
```

```
ceci_est_un_objet
```

```
[1] 11.2
```

5.1.d Création séquence de nombre

Pour afficher la page d'aide, il faut utiliser `?seq`

```
seq(from = 1, to = 50, by = 5)
```

```
 [1]  1  6 11 16 21 26 31 36 41 46
```

5.2 Structure et type des données

5.2.a Nature des données

```
class(iris)
```

```
[1] "data.frame"
```

```
class(iris$Species)
```

```
[1] "factor"
```

Le signe `$` permet d'accéder à une colonne ou un élément de liste.

```
class(iris$Petal.Width)
```

```
[1] "numeric"
```

5.2.b Structure des données

création d'un vecteur

```
(vecteur <- c(1:10))
```

```
 [1]  1  2  3  4  5  6  7  8  9 10
```

création d'une matrice

```
(matrice <- matrix(data = 1:10, ncol = 2))
```

```
     [,1] [,2]
[1,]    1    6
[2,]    2    7
[3,]    3    8
[4,]    4    9
[5,]    5   10
```

création d'un tableau de données

```
(tableau_de_donnees <- data.frame(vecteur, matrice))
```

```
   vecteur X1 X2
1        1  1  6
2        2  2  7
3        3  3  8
4        4  4  9
5        5  5 10
6        6  1  6
7        7  2  7
8        8  3  8
9        9  4  9
10      10  5 10
```

création d'une liste

```
(liste <- list(vecteur, LETTERS[1:5], tableau_de_donnees))
```

```
[[1]]
 [1]  1  2  3  4  5  6  7  8  9 10
```

```
[[2]]
[1] "A" "B" "C" "D" "E"

[[3]]
   vecteur X1 X2
1        1  1  6
2        2  2  7
3        3  3  8
4        4  4  9
5        5  5 10
6        6  1  6
7        7  2  7
8        8  3  8
9        9  4  9
10      10  5 10
```

5.3 Importer des données

5.3.a Jeux de données disponibles dans R

```
data()
```

5.3.b Format **csv**

Création d'un fichier csv

```
write.csv(
  mtcars,
  "data/jdd_voiture.csv"
)
```

lecture d'un fichier csv

```
jdd_voiture <- read.csv(
  file = "data/jdd_voiture.csv",
  row.names = "X"
)
```

visualisation du jeu de données

```
tibble::view(jdd_voiture)
```

création d'un résumé du jeu de données

```r
summary(jdd_voiture)
```

```
      mpg             cyl             disp             hp       
 Min.   :10.40   Min.   :4.000   Min.   : 71.1   Min.   : 52.0  
 1st Qu.:15.43   1st Qu.:4.000   1st Qu.:120.8   1st Qu.: 96.5  
 Median :19.20   Median :6.000   Median :196.3   Median :123.0  
 Mean   :20.09   Mean   :6.188   Mean   :230.7   Mean   :146.7  
 3rd Qu.:22.80   3rd Qu.:8.000   3rd Qu.:326.0   3rd Qu.:180.0  
 Max.   :33.90   Max.   :8.000   Max.   :472.0   Max.   :335.0  
      drat             wt             qsec             vs        
 Min.   :2.760   Min.   :1.513   Min.   :14.50   Min.   :0.0000  
 1st Qu.:3.080   1st Qu.:2.581   1st Qu.:16.89   1st Qu.:0.0000  
 Median :3.695   Median :3.325   Median :17.71   Median :0.0000  
 Mean   :3.597   Mean   :3.217   Mean   :17.85   Mean   :0.4375  
 3rd Qu.:3.920   3rd Qu.:3.610   3rd Qu.:18.90   3rd Qu.:1.0000  
 Max.   :4.930   Max.   :5.424   Max.   :22.90   Max.   :1.0000  
       am              gear            carb      
 Min.   :0.0000   Min.   :3.000   Min.   :1.000  
 1st Qu.:0.0000   1st Qu.:3.000   1st Qu.:2.000  
 Median :0.0000   Median :4.000   Median :2.000  
 Mean   :0.4062   Mean   :3.688   Mean   :2.812  
 3rd Qu.:1.0000   3rd Qu.:4.000   3rd Qu.:4.000  
 Max.   :1.0000   Max.   :5.000   Max.   :8.000  
```

5.3.c Format `xlsx`

creation d'un fichier excel

```r
library(openxlsx)

write.xlsx(
  women,
  "data/jdd_femme.xlsx"
)
```

lire un fichier excel

```r
jdd_femme <- read.xlsx(
  "data/jdd_femme.xlsx"
)
```

5.3.d Format `txt`

Lecture de données depuis un site internet

```r
library(rvest)
page <-  read_html("https://fr.m.wikisource.org/wiki/Lettres_de_la_comtesse_de_
S%C3%A9gur/Texte_entier") |>
  html_nodes("h1") |>
  html_text()
```

Enregistrement dans un fichier `txt`

```r
write(page, "data/titre_segur.txt")
# ou
page |>
  write("data/titre_segur.txt")
```

lecture d'un fichier txt

```r
titre_segur <- read.delim(
  "data/titre_segur.txt",
  header = FALSE
)
```

5.4 utilisation du pipe natif

sans le pipe, en créant des objets

```r
# création d'objets intermédiaire
etape1 <- c(1:30)
etape2 <- sqrt(etape1)
etape3 <- mean(etape2)
floor(etape3)
```

```
[1] 3
```

```r
# sans pipe
floor(mean(sqrt(c(1:10)*3)))
```

```
[1] 3
```

```r
# équivalent à
floor(
  mean(
    sqrt(
```

```r
      c(1:30)
    )
  )
)
```

```
[1] 3
```

```r
# avec pipe
c(1:30) |>
  sqrt() |>
  mean() |>
  floor()
```

```
[1] 3
```

6 Travailler sur des jeux de données

6.1 Manipulation des données dans R

6.1.a Sélection ligne(s) et/ou colonne(s)

langage de base

```r
(exemple <- matrix(data = c(12, 53, 6, 8, 97, 42), nrow = 3))
```

```
     [,1] [,2]
[1,]   12    8
[2,]   53   97
[3,]    6   42
```

```r
exemple[3,2]
```

```
[1] 42
```

```r
colnames(exemple) <- list("premiere_colonne", "deuxieme_colonne")
rownames(exemple) <- c("premiere_ligne", "deuxieme_ligne", "troisieme_ligne")
exemple
```

```
                 premiere_colonne deuxieme_colonne
premiere_ligne                 12                8
deuxieme_ligne                 53               97
troisieme_ligne                 6               42
```

```r
exemple["troisieme_ligne", "deuxieme_colonne"]
```

```
[1] 42
```

exemple en {tidyverse}

```r
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.2.1     ✔ readr     2.2.0
✔ forcats   1.0.1     ✔ stringr   1.6.0
✔ ggplot2   4.0.3     ✔ tibble    3.3.1
✔ lubridate 1.9.5     ✔ tidyr     1.3.2
✔ purrr     1.2.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts()
──
✖ dplyr::filter()         masks stats::filter()
✖ readr::guess_encoding() masks rvest::guess_encoding()
✖ dplyr::lag()            masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```r
(exemple_tibble <- tibble(
  premiere_colonne = c(12, 53, 6),
  deuxieme_colonne = c(8, 97, 42)
))
```

```
# A tibble: 3 × 2
  premiere_colonne deuxieme_colonne
             <dbl>            <dbl>
1               12                8
2               53               97
3                6               42
```

```
exemple_tibble[3,2]
```

```
# A tibble: 1 × 1
  deuxieme_colonne
             <dbl>
1               42
```

```
exemple_tibble[3, "deuxieme_colonne"]
```

```
# A tibble: 1 × 1
  deuxieme_colonne
             <dbl>
1               42
```

6.1.b Sélection de plusieurs lignes ou colonnes

```
exemple[c(1,3),]
```

```
                premiere_colonne deuxieme_colonne
premiere_ligne                12                8
troisieme_ligne                6               42
```

```
exemple_tibble[c(1:2), 1]
```

```
# A tibble: 2 × 1
  premiere_colonne
             <dbl>
1               12
2               53
```

6.1.c Utilisation des parenthèses autours du code

```
test_affichage <- "Ceci est un test pour montrer l'utilisation des parenthèses"
(test_affichage <- "Ceci est un test pour montrer l'utilisation des parenthèses")
```

```
[1] "Ceci est un test pour montrer l'utilisation des parenthèses"
```

6.1.d Choix de lignes et colonnes

```
iris[iris$Species == "setosa", ]
```

```
   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1           5.1         3.5          1.4         0.2  setosa
2           4.9         3.0          1.4         0.2  setosa
3           4.7         3.2          1.3         0.2  setosa
4           4.6         3.1          1.5         0.2  setosa
5           5.0         3.6          1.4         0.2  setosa
6           5.4         3.9          1.7         0.4  setosa
7           4.6         3.4          1.4         0.3  setosa
8           5.0         3.4          1.5         0.2  setosa
9           4.4         2.9          1.4         0.2  setosa
10          4.9         3.1          1.5         0.1  setosa
11          5.4         3.7          1.5         0.2  setosa
12          4.8         3.4          1.6         0.2  setosa
13          4.8         3.0          1.4         0.1  setosa
14          4.3         3.0          1.1         0.1  setosa
15          5.8         4.0          1.2         0.2  setosa
16          5.7         4.4          1.5         0.4  setosa
17          5.4         3.9          1.3         0.4  setosa
18          5.1         3.5          1.4         0.3  setosa
19          5.7         3.8          1.7         0.3  setosa
20          5.1         3.8          1.5         0.3  setosa
21          5.4         3.4          1.7         0.2  setosa
22          5.1         3.7          1.5         0.4  setosa
23          4.6         3.6          1.0         0.2  setosa
24          5.1         3.3          1.7         0.5  setosa
25          4.8         3.4          1.9         0.2  setosa
26          5.0         3.0          1.6         0.2  setosa
27          5.0         3.4          1.6         0.4  setosa
28          5.2         3.5          1.5         0.2  setosa
29          5.2         3.4          1.4         0.2  setosa
30          4.7         3.2          1.6         0.2  setosa
31          4.8         3.1          1.6         0.2  setosa
32          5.4         3.4          1.5         0.4  setosa
33          5.2         4.1          1.5         0.1  setosa
34          5.5         4.2          1.4         0.2  setosa
35          4.9         3.1          1.5         0.2  setosa
36          5.0         3.2          1.2         0.2  setosa
37          5.5         3.5          1.3         0.2  setosa
38          4.9         3.6          1.4         0.1  setosa
```

```
39          4.4         3.0          1.3         0.2  setosa
40          5.1         3.4          1.5         0.2  setosa
41          5.0         3.5          1.3         0.3  setosa
42          4.5         2.3          1.3         0.3  setosa
43          4.4         3.2          1.3         0.2  setosa
44          5.0         3.5          1.6         0.6  setosa
45          5.1         3.8          1.9         0.4  setosa
46          4.8         3.0          1.4         0.3  setosa
47          5.1         3.8          1.6         0.2  setosa
48          4.6         3.2          1.4         0.2  setosa
49          5.3         3.7          1.5         0.2  setosa
50          5.0         3.3          1.4         0.2  setosa
```

```
iris[iris$Species == "setosa", "Sepal.Length"]
```

```
 [1] 5.1 4.9 4.7 4.6 5.0 5.4 4.6 5.0 4.4 4.9 5.4 4.8 4.8 4.3 5.8 5.7 5.4 5.1 5.7
[20] 5.1 5.4 5.1 4.6 5.1 4.8 5.0 5.0 5.2 5.2 4.7 4.8 5.4 5.2 5.5 4.9 5.0 5.5 4.9
[39] 4.4 5.1 5.0 4.5 4.4 5.0 5.1 4.8 5.1 4.6 5.3 5.0
```

```
iris[iris$Species == "setosa" |  iris$Sepal.Length < 5, ]
```

```
    Sepal.Length Sepal.Width Petal.Length Petal.Width    Species
1            5.1         3.5          1.4         0.2     setosa
2            4.9         3.0          1.4         0.2     setosa
3            4.7         3.2          1.3         0.2     setosa
4            4.6         3.1          1.5         0.2     setosa
5            5.0         3.6          1.4         0.2     setosa
6            5.4         3.9          1.7         0.4     setosa
7            4.6         3.4          1.4         0.3     setosa
8            5.0         3.4          1.5         0.2     setosa
9            4.4         2.9          1.4         0.2     setosa
10           4.9         3.1          1.5         0.1     setosa
11           5.4         3.7          1.5         0.2     setosa
12           4.8         3.4          1.6         0.2     setosa
13           4.8         3.0          1.4         0.1     setosa
14           4.3         3.0          1.1         0.1     setosa
15           5.8         4.0          1.2         0.2     setosa
16           5.7         4.4          1.5         0.4     setosa
17           5.4         3.9          1.3         0.4     setosa
18           5.1         3.5          1.4         0.3     setosa
19           5.7         3.8          1.7         0.3     setosa
```

```
20           5.1         3.8          1.5         0.3     setosa
21           5.4         3.4          1.7         0.2     setosa
22           5.1         3.7          1.5         0.4     setosa
23           4.6         3.6          1.0         0.2     setosa
24           5.1         3.3          1.7         0.5     setosa
25           4.8         3.4          1.9         0.2     setosa
26           5.0         3.0          1.6         0.2     setosa
27           5.0         3.4          1.6         0.4     setosa
28           5.2         3.5          1.5         0.2     setosa
29           5.2         3.4          1.4         0.2     setosa
30           4.7         3.2          1.6         0.2     setosa
31           4.8         3.1          1.6         0.2     setosa
32           5.4         3.4          1.5         0.4     setosa
33           5.2         4.1          1.5         0.1     setosa
34           5.5         4.2          1.4         0.2     setosa
35           4.9         3.1          1.5         0.2     setosa
36           5.0         3.2          1.2         0.2     setosa
37           5.5         3.5          1.3         0.2     setosa
38           4.9         3.6          1.4         0.1     setosa
39           4.4         3.0          1.3         0.2     setosa
40           5.1         3.4          1.5         0.2     setosa
41           5.0         3.5          1.3         0.3     setosa
42           4.5         2.3          1.3         0.3     setosa
43           4.4         3.2          1.3         0.2     setosa
44           5.0         3.5          1.6         0.6     setosa
45           5.1         3.8          1.9         0.4     setosa
46           4.8         3.0          1.4         0.3     setosa
47           5.1         3.8          1.6         0.2     setosa
48           4.6         3.2          1.4         0.2     setosa
49           5.3         3.7          1.5         0.2     setosa
50           5.0         3.3          1.4         0.2     setosa
58           4.9         2.4          3.3         1.0 versicolor
107          4.9         2.5          4.5         1.7  virginica
```

```
iris[iris$Species == "setosa" & iris$Sepal.Length < 5, ]
```

```
   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
2           4.9         3.0          1.4         0.2  setosa
3           4.7         3.2          1.3         0.2  setosa
4           4.6         3.1          1.5         0.2  setosa
7           4.6         3.4          1.4         0.3  setosa
9           4.4         2.9          1.4         0.2  setosa
```

```
10          4.9         3.1          1.5         0.1  setosa
12          4.8         3.4          1.6         0.2  setosa
13          4.8         3.0          1.4         0.1  setosa
14          4.3         3.0          1.1         0.1  setosa
23          4.6         3.6          1.0         0.2  setosa
25          4.8         3.4          1.9         0.2  setosa
30          4.7         3.2          1.6         0.2  setosa
31          4.8         3.1          1.6         0.2  setosa
35          4.9         3.1          1.5         0.2  setosa
38          4.9         3.6          1.4         0.1  setosa
39          4.4         3.0          1.3         0.2  setosa
42          4.5         2.3          1.3         0.3  setosa
43          4.4         3.2          1.3         0.2  setosa
46          4.8         3.0          1.4         0.3  setosa
48          4.6         3.2          1.4         0.2  setosa
```

```
filter(iris, Species == "setosa")
```

```
   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1           5.1         3.5          1.4         0.2  setosa
2           4.9         3.0          1.4         0.2  setosa
3           4.7         3.2          1.3         0.2  setosa
4           4.6         3.1          1.5         0.2  setosa
5           5.0         3.6          1.4         0.2  setosa
6           5.4         3.9          1.7         0.4  setosa
7           4.6         3.4          1.4         0.3  setosa
8           5.0         3.4          1.5         0.2  setosa
9           4.4         2.9          1.4         0.2  setosa
10          4.9         3.1          1.5         0.1  setosa
11          5.4         3.7          1.5         0.2  setosa
12          4.8         3.4          1.6         0.2  setosa
13          4.8         3.0          1.4         0.1  setosa
14          4.3         3.0          1.1         0.1  setosa
15          5.8         4.0          1.2         0.2  setosa
16          5.7         4.4          1.5         0.4  setosa
17          5.4         3.9          1.3         0.4  setosa
18          5.1         3.5          1.4         0.3  setosa
19          5.7         3.8          1.7         0.3  setosa
20          5.1         3.8          1.5         0.3  setosa
21          5.4         3.4          1.7         0.2  setosa
22          5.1         3.7          1.5         0.4  setosa
23          4.6         3.6          1.0         0.2  setosa
```

```
24          5.1         3.3         1.7         0.5  setosa
25          4.8         3.4         1.9         0.2  setosa
26          5.0         3.0         1.6         0.2  setosa
27          5.0         3.4         1.6         0.4  setosa
28          5.2         3.5         1.5         0.2  setosa
29          5.2         3.4         1.4         0.2  setosa
30          4.7         3.2         1.6         0.2  setosa
31          4.8         3.1         1.6         0.2  setosa
32          5.4         3.4         1.5         0.4  setosa
33          5.2         4.1         1.5         0.1  setosa
34          5.5         4.2         1.4         0.2  setosa
35          4.9         3.1         1.5         0.2  setosa
36          5.0         3.2         1.2         0.2  setosa
37          5.5         3.5         1.3         0.2  setosa
38          4.9         3.6         1.4         0.1  setosa
39          4.4         3.0         1.3         0.2  setosa
40          5.1         3.4         1.5         0.2  setosa
41          5.0         3.5         1.3         0.3  setosa
42          4.5         2.3         1.3         0.3  setosa
43          4.4         3.2         1.3         0.2  setosa
44          5.0         3.5         1.6         0.6  setosa
45          5.1         3.8         1.9         0.4  setosa
46          4.8         3.0         1.4         0.3  setosa
47          5.1         3.8         1.6         0.2  setosa
48          4.6         3.2         1.4         0.2  setosa
49          5.3         3.7         1.5         0.2  setosa
50          5.0         3.3         1.4         0.2  setosa
```

```
filter(iris, Species == "setosa") |> select(Sepal.Length)
```

```
   Sepal.Length
1           5.1
2           4.9
3           4.7
4           4.6
5           5.0
6           5.4
7           4.6
8           5.0
9           4.4
10          4.9
11          5.4
```

```
12          4.8
13          4.8
14          4.3
15          5.8
16          5.7
17          5.4
18          5.1
19          5.7
20          5.1
21          5.4
22          5.1
23          4.6
24          5.1
25          4.8
26          5.0
27          5.0
28          5.2
29          5.2
30          4.7
31          4.8
32          5.4
33          5.2
34          5.5
35          4.9
36          5.0
37          5.5
38          4.9
39          4.4
40          5.1
41          5.0
42          4.5
43          4.4
44          5.0
45          5.1
46          4.8
47          5.1
48          4.6
49          5.3
50          5.0
```

```
filter(iris, Species == "setosa" | iris$Sepal.Length < 5)
```

```
   Sepal.Length Sepal.Width Petal.Length Petal.Width    Species
1           5.1         3.5          1.4         0.2     setosa
2           4.9         3.0          1.4         0.2     setosa
3           4.7         3.2          1.3         0.2     setosa
4           4.6         3.1          1.5         0.2     setosa
5           5.0         3.6          1.4         0.2     setosa
6           5.4         3.9          1.7         0.4     setosa
7           4.6         3.4          1.4         0.3     setosa
8           5.0         3.4          1.5         0.2     setosa
9           4.4         2.9          1.4         0.2     setosa
10          4.9         3.1          1.5         0.1     setosa
11          5.4         3.7          1.5         0.2     setosa
12          4.8         3.4          1.6         0.2     setosa
13          4.8         3.0          1.4         0.1     setosa
14          4.3         3.0          1.1         0.1     setosa
15          5.8         4.0          1.2         0.2     setosa
16          5.7         4.4          1.5         0.4     setosa
17          5.4         3.9          1.3         0.4     setosa
18          5.1         3.5          1.4         0.3     setosa
19          5.7         3.8          1.7         0.3     setosa
20          5.1         3.8          1.5         0.3     setosa
21          5.4         3.4          1.7         0.2     setosa
22          5.1         3.7          1.5         0.4     setosa
23          4.6         3.6          1.0         0.2     setosa
24          5.1         3.3          1.7         0.5     setosa
25          4.8         3.4          1.9         0.2     setosa
26          5.0         3.0          1.6         0.2     setosa
27          5.0         3.4          1.6         0.4     setosa
28          5.2         3.5          1.5         0.2     setosa
29          5.2         3.4          1.4         0.2     setosa
30          4.7         3.2          1.6         0.2     setosa
31          4.8         3.1          1.6         0.2     setosa
32          5.4         3.4          1.5         0.4     setosa
33          5.2         4.1          1.5         0.1     setosa
34          5.5         4.2          1.4         0.2     setosa
35          4.9         3.1          1.5         0.2     setosa
36          5.0         3.2          1.2         0.2     setosa
37          5.5         3.5          1.3         0.2     setosa
38          4.9         3.6          1.4         0.1     setosa
39          4.4         3.0          1.3         0.2     setosa
40          5.1         3.4          1.5         0.2     setosa
41          5.0         3.5          1.3         0.3     setosa
42          4.5         2.3          1.3         0.3     setosa
```

```
43          4.4         3.2          1.3         0.2     setosa
44          5.0         3.5          1.6         0.6     setosa
45          5.1         3.8          1.9         0.4     setosa
46          4.8         3.0          1.4         0.3     setosa
47          5.1         3.8          1.6         0.2     setosa
48          4.6         3.2          1.4         0.2     setosa
49          5.3         3.7          1.5         0.2     setosa
50          5.0         3.3          1.4         0.2     setosa
51          4.9         2.4          3.3         1.0 versicolor
52          4.9         2.5          4.5         1.7  virginica
```

```
filter(iris, Species == "setosa" & iris$Sepal.Length < 5)
```

```
   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1           4.9         3.0          1.4         0.2  setosa
2           4.7         3.2          1.3         0.2  setosa
3           4.6         3.1          1.5         0.2  setosa
4           4.6         3.4          1.4         0.3  setosa
5           4.4         2.9          1.4         0.2  setosa
6           4.9         3.1          1.5         0.1  setosa
7           4.8         3.4          1.6         0.2  setosa
8           4.8         3.0          1.4         0.1  setosa
9           4.3         3.0          1.1         0.1  setosa
10          4.6         3.6          1.0         0.2  setosa
11          4.8         3.4          1.9         0.2  setosa
12          4.7         3.2          1.6         0.2  setosa
13          4.8         3.1          1.6         0.2  setosa
14          4.9         3.1          1.5         0.2  setosa
15          4.9         3.6          1.4         0.1  setosa
16          4.4         3.0          1.3         0.2  setosa
17          4.5         2.3          1.3         0.3  setosa
18          4.4         3.2          1.3         0.2  setosa
19          4.8         3.0          1.4         0.3  setosa
20          4.6         3.2          1.4         0.2  setosa
```

6.2 Jointure

6.2.a création des tables

```
(table_gauche <- tibble(
  id = LETTERS[1:3],
  x = c(5, 9, 7),
```

```
  y = c(1, 4, 8)
))
```

```
# A tibble: 3 × 3
  id        x     y
  <chr> <dbl> <dbl>
1 A         5     1
2 B         9     4
3 C         7     8
```

```
(table_droite <- tibble(
  identite = LETTERS[c(1, 2, 4)],
  z = c(4, 7, 6),
  w = c(2, 8, 6)
))
```

```
# A tibble: 3 × 3
  identite     z     w
  <chr>    <dbl> <dbl>
1 A            4     2
2 B            7     8
3 D            6     6
```

6.2.b jointures

```
inner_join(
  table_gauche,
  table_droite,
  by = c("id" = "identite")
)
```

```
# A tibble: 2 × 5
  id        x     y     z     w
  <chr> <dbl> <dbl> <dbl> <dbl>
1 A         5     1     4     2
2 B         9     4     7     8
```

```
full_join(
  table_gauche,
  table_droite,
```

```
  by = c("id" = "identite")
)
```

```
# A tibble: 4 × 5
  id        x     y     z     w
  <chr> <dbl> <dbl> <dbl> <dbl>
1 A         5     1     4     2
2 B         9     4     7     8
3 C         7     8    NA    NA
4 D        NA    NA     6     6
```

```
left_join(
  table_gauche,
  table_droite,
  by = c("id" = "identite")
)
```

```
# A tibble: 3 × 5
  id        x     y     z     w
  <chr> <dbl> <dbl> <dbl> <dbl>
1 A         5     1     4     2
2 B         9     4     7     8
3 C         7     8    NA    NA
```

```
right_join(
  table_gauche,
  table_droite,
  by = c("id" = "identite")
)
```

```
# A tibble: 3 × 5
  id        x     y     z     w
  <chr> <dbl> <dbl> <dbl> <dbl>
1 A         5     1     4     2
2 B         9     4     7     8
3 D        NA    NA     6     6
```

6.2.c vérification

```
summary(table_gauche)
```

```
       id              x               y        
 Length   :3   Min.   :5   Min.   :1.000  
 N.unique :3   1st Qu.:6   1st Qu.:2.500  
 N.blank  :0   Median :7   Median :4.000  
 Min.nchar:1   Mean   :7   Mean   :4.333  
 Max.nchar:1   3rd Qu.:8   3rd Qu.:6.000  
               Max.   :9   Max.   :8.000  
```

```
summary(table_droite)
```

```
     identite           z               w        
 Length   :3   Min.   :4.000   Min.   :2.000  
 N.unique :3   1st Qu.:5.000   1st Qu.:4.000  
 N.blank  :0   Median :6.000   Median :6.000  
 Min.nchar:1   Mean   :5.667   Mean   :5.333  
 Max.nchar:1   3rd Qu.:6.500   3rd Qu.:7.000  
               Max.   :7.000   Max.   :8.000  
```

```
table_gauche |>
  count(id) |>
  filter(n > 1)
```

```
# A tibble: 0 × 2
# ℹ 2 variables: id <chr>, n <int>
```

```
table_droite |>
  count(identite) |>
  filter(n > 1)
```

```
# A tibble: 0 × 2
# ℹ 2 variables: identite <chr>, n <int>
```

```
table_gauche$id |> class()
```

```
[1] "character"
```

```
table_droite$identite |> class()
```

```
[1] "character"
```

```
nrow(table_gauche)
```

```
[1] 3
```

```
nrow(table_droite)
```

```
[1] 3
```

```
table_gauche$id %in% table_droite$identite |> sum()
```

```
[1] 2
```